SESD-Assignment-2

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Case Study:
A **Smart Parking System** helps drivers find and reserve parking spots via a mobile app. Drivers can check spot availability, make payments, and receive directions. Admins manage spot inventory and monitor usage analytics. The system includes sensors to detect car presence and notifies users of real-time changes.

Tasks:

- **Use Case Diagram**: Identify actors like Driver, Admin, Sensor; include use cases for booking, monitoring, payment, and alerts.
- **Class Diagram**: Model entities like ParkingSpot, Sensor, User, Payment, and Reservation.
- **Sequence Diagram**: Model the process of reserving a spot and completing payment.
- **Activity Diagram**: Show the workflow from searching to exiting the parking lot.
- **State Machine Diagram**: Represent ParkingSpot states: Available → Reserved → Occupied → Available.

1. Aim

To design a Smart Parking System using UML diagrams that allows drivers to search, reserve, and pay for parking spots while enabling administrators to manage parking spaces and monitor their usage.

2. Case Study Description

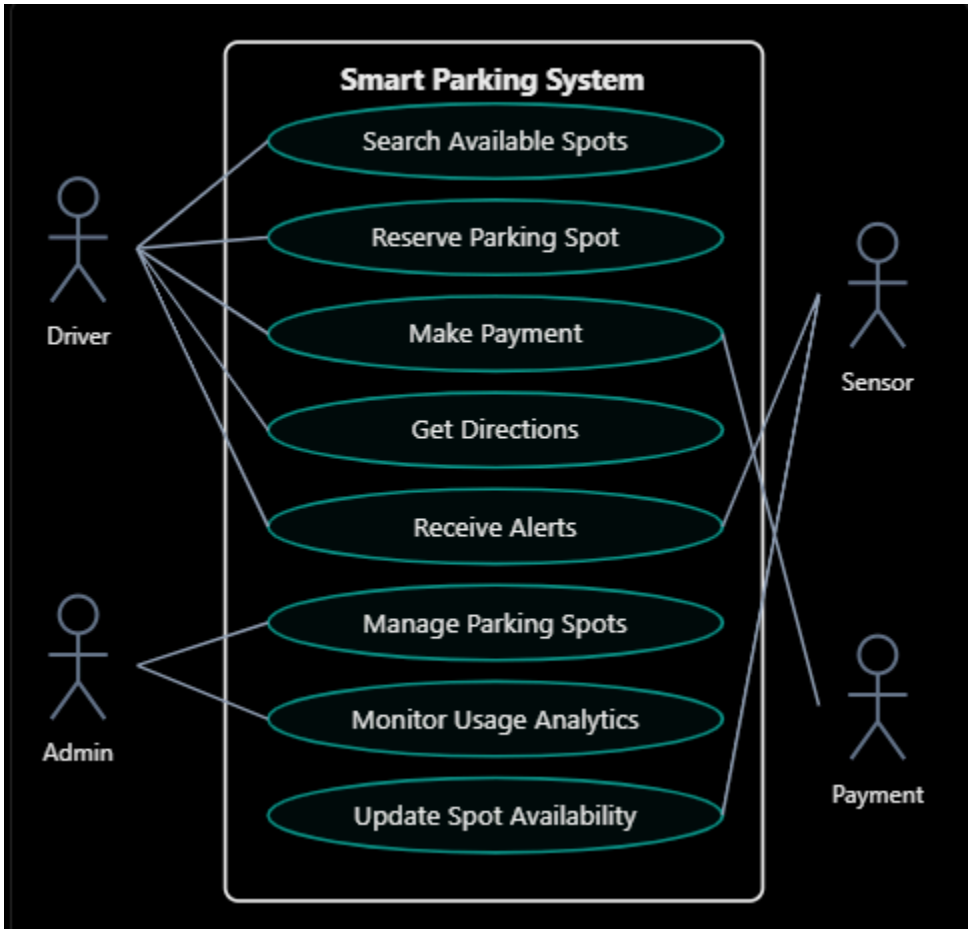
The Smart Parking System is a mobile application that helps drivers find available parking spots, reserve them, make payments, and receive directions.

The system uses sensors to detect vehicle presence and update parking availability in real time. Administrators manage parking spots and monitor parking usage.

3. Use Case Diagram

Objective: To represent the interactions between the actors and the Smart Parking System.

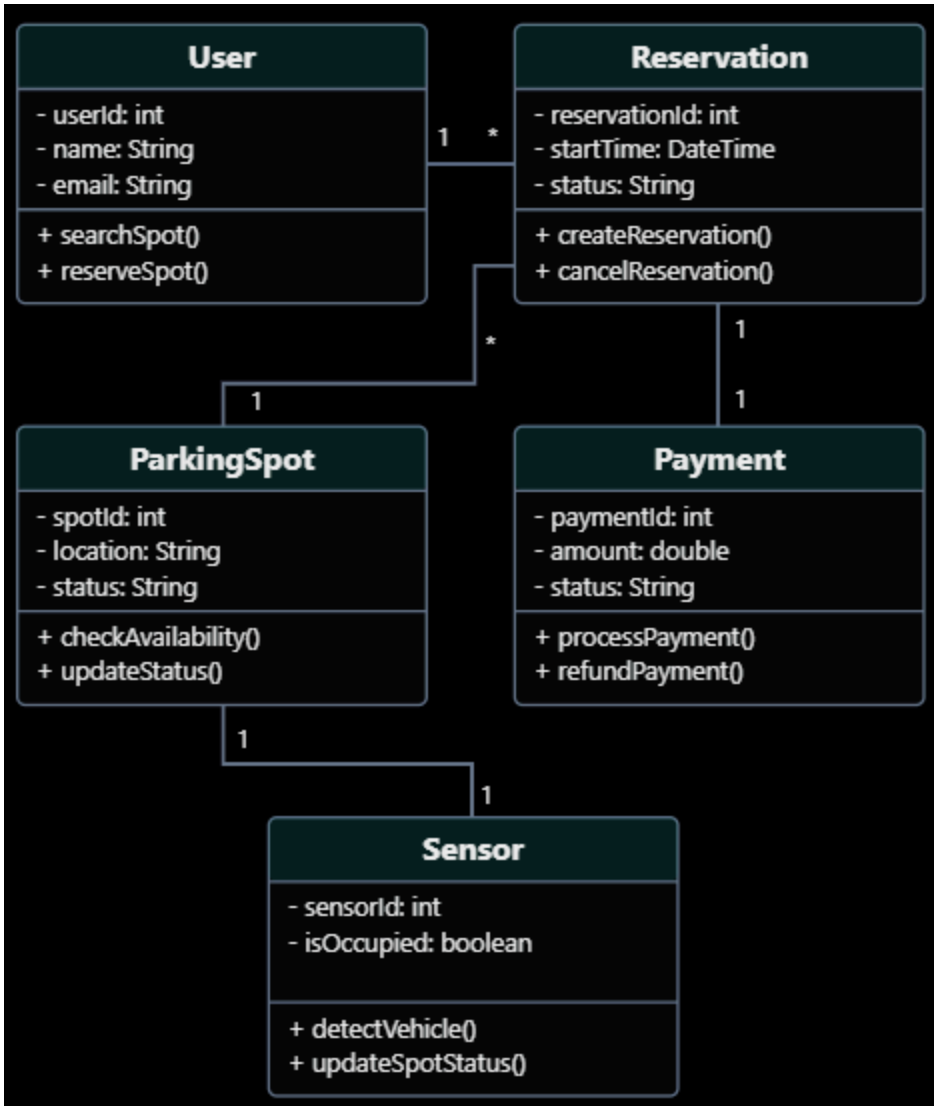
Actors: Driver, Admin, Sensor, Payment Gateway.



Explanation: The Driver searches, reserves, pays, and receives notifications. The Admin manages parking spots and usage reports. Sensors update spot availability, while the Payment Gateway processes payments.

4. Class Diagram

Objective: To represent the classes, attributes, methods, and relationships of the system.

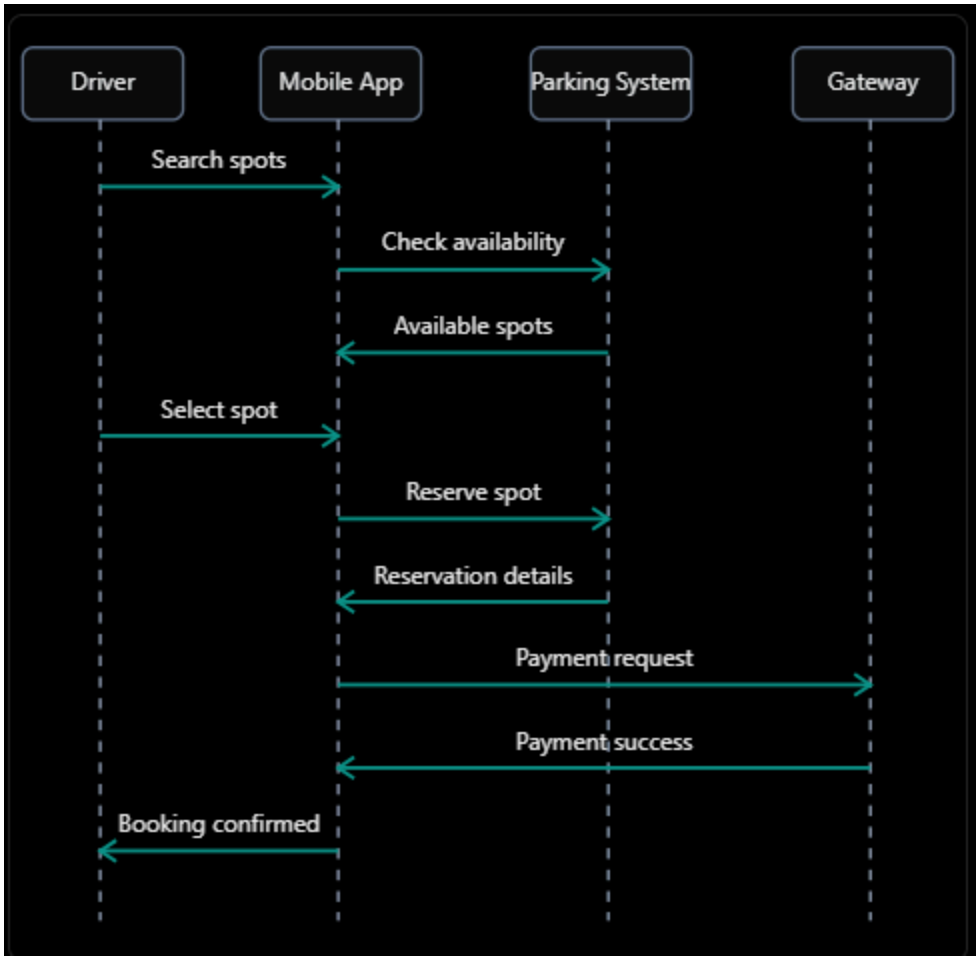


Class relationships:

- One User can make multiple Reservations.
- Each Reservation is associated with one ParkingSpot and one Payment.
- Each ParkingSpot has one Sensor to detect vehicle occupancy.

5. Sequence Diagram

Objective: To show the sequence of interactions during parking reservation and payment.

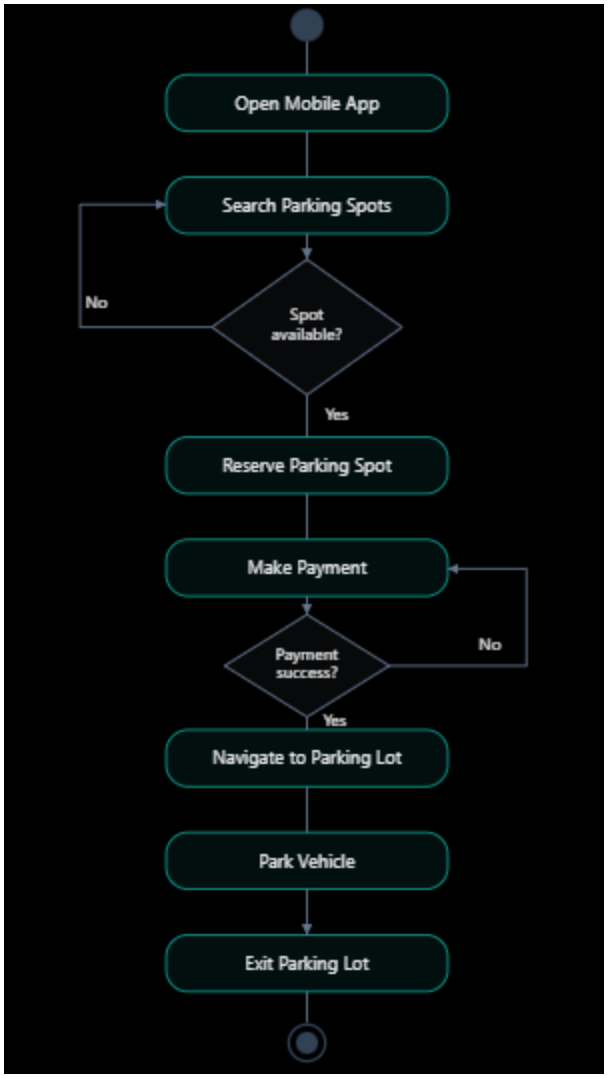


Explanation:

1. Driver searches for available parking spots.
2. System displays available spots.
3. Driver selects a spot and requests a reservation.
4. System temporarily holds the spot and generates payment details.
5. Driver completes payment through the Payment Gateway.
6. After successful payment, the system confirms the reservation and notifies the driver.

6. Activity Diagram

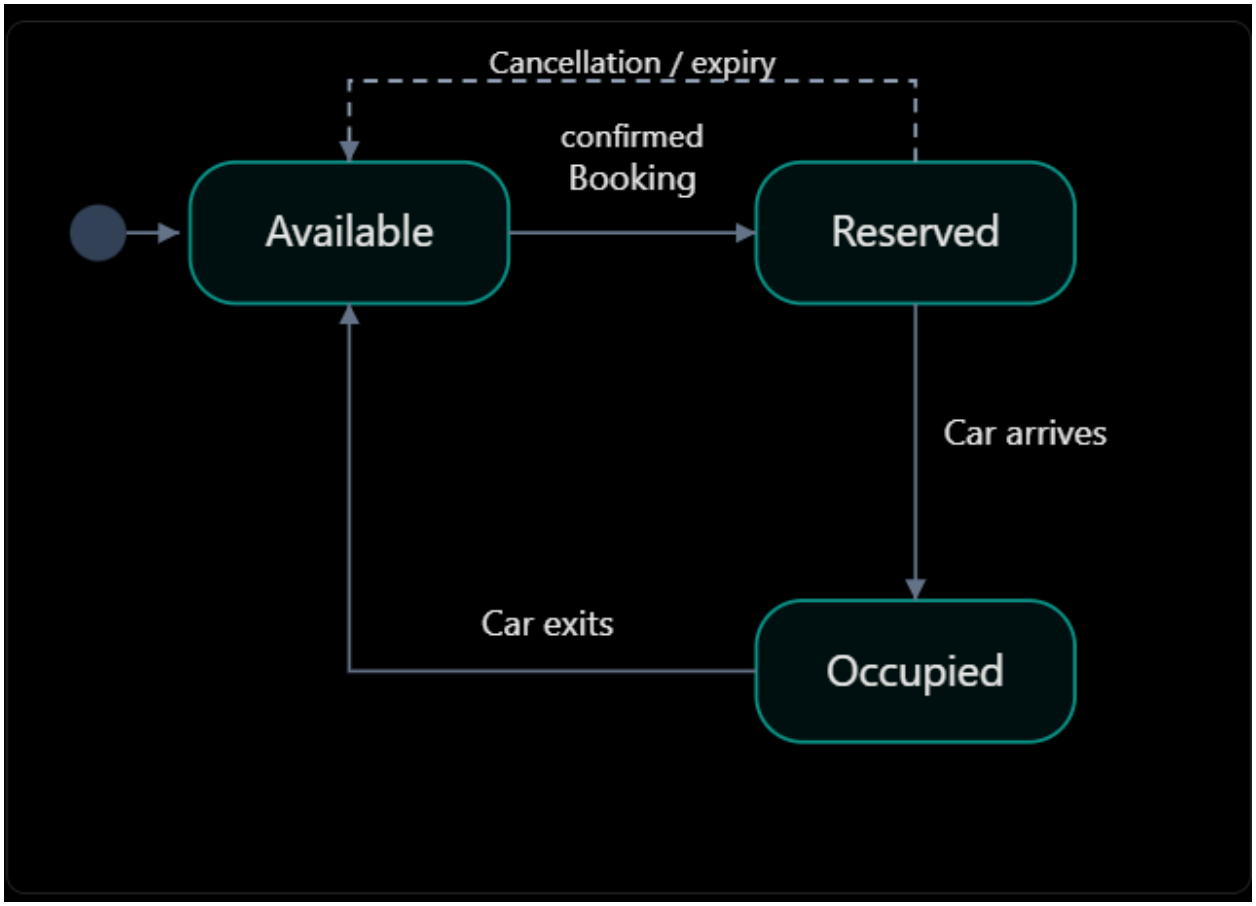
Objective: To represent the complete workflow from searching for a parking spot to exiting the parking lot.



Explanation: The process begins when a driver opens the app and searches for a parking spot. If a spot is available, the driver reserves it and makes payment. After successful payment, the driver receives directions, parks the vehicle, and finally exits. The sensor then updates the spot as available.

7. State Machine Diagram

Objective: To represent the different states of a ParkingSpot and transitions between them.



State descriptions:

- Available: Parking spot is empty and ready for booking.
- Reserved: A driver has successfully booked the parking spot.
- Occupied: The vehicle has arrived and is detected by the sensor.
- Available: When the vehicle exits, the sensor detects the empty spot and updates its status.

A cancelled or expired reservation also returns the spot to Available if it is not occupied.

8. Conclusion

The Smart Parking System is designed using five UML diagrams: Use Case, Class, Sequence, Activity, and State Machine. These diagrams explain the system's functionality, structure, interactions, workflow, and parking spot states. The system helps drivers save time, enables secure parking reservations and payments, and improves parking management through real-time sensor updates.